

upGrad

Data foundations: from S3 objects to analysable tables

M00 Session 02

Session map

01 Give data a readable shape

Distinguish storage from a trusted, documented data contract.

02 Validate before analysis

Inspect types, missingness, time and entity keys before transformations begin.

Each topic can be compiled alone while this session file assembles the complete learner deck.

TOPIC 01

Give data a readable shape

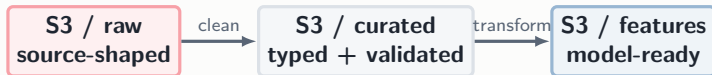
Distinguish storage from a trusted, documented data contract.

What this topic establishes

By the end of this topic, learners can

- explain why an S3 prefix is not a database table; and
- state a simple rule for separating raw, curated and derived data.

A data lake needs a readable convention



Definition – Data contract

An agreed description of fields, types, owners, freshness and allowed values. It lets downstream work fail clearly instead of silently drifting.

TOPIC 02

Validate before analysis

Inspect types, missingness, time and entity keys before transformations begin.

Inspect before you transform

```
1 import pandas as pd
2
3 orders = pd.read_csv("data/orders.csv")
4 print(orders.dtypes)
5 print(orders.isna().mean().sort_values(ascending=False).head())
6
7 orders["ordered_at"] = pd.to_datetime(orders["ordered_at"], utc=True)
8 assert orders["order_id"].is_unique
9
```

The first model feature is often a validation check: are we measuring the entity and time period we think we are?

Key takeaway

Key takeaway – Storage is not semantics

S3 gives durable object storage. Naming conventions, schema checks and documented transformations make those objects safe to analyse.